

ESS INDIA Handover Document

ESS INDIA - Project Handover Document

Technical & Operational Handover

Prepared For

ESS India Client

Prepared By

SJ Media Labs

Document Version

v1.0

Date

July 2026

Document Control

Item	Details
Project	ESS India
Document	Technical & Operational Handover
Version	1.0
Status	Final
Prepared By	SJ Media Labs
Reviewed By	_____
Approved By	_____
Handover Date	_____

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1. Executive Summary

Project Overview

The **ESS India Portal** is an enterprise-grade digital website and custom Content Management System (CMS) designed using modern cloud-native technologies. The application has been engineered to provide superior page loading speeds, high security, complete administrative autonomy, and seamless future scalability.

The solution consists of three major components:

1. **Customer Website:** A fully responsive, modern web front-end optimized for search engines (SEO) and user engagement.
 2. **Administration Portal:** A robust dashboard for managers and editors to modify pages, build layouts, edit menus, view analytics, and manage career applications.
 3. **REST API Backend:** A secure server-side logic layer for page rendering, forms capture, notifications dispatch, and asset uploads.
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2. Solution Overview

Customer Portal

- **Home Page:** High-impact introduction page featuring dynamic carousels, corporate values, and service pathways.
- **Custom Pages:** Dynamically created standard layouts (e.g. About, Spotlights, Case Studies, Contact).
- **Careers Module:** View listings, search positions, view detailed JDs, and submit applications (with resume upload).
- **Inquiry / Lead Forms:** Country-code specific phone inputs and validated email forms capturing customer outreach.
- **Dynamic Menu & Navigation:** Responsive multi-behavior navigation (Standard Links, Dropdowns, and Mega Menus).

Administration Portal

- **Dashboard Panel:** Key performance indicators showing page counts, recent applications, and content revisions.
- **Pages Builder Wizard:** Full-width wizard interface to create pages, configure slugs, edit section contents, and manage SEO parameters.
- **Navigation Tree Manager:** Drag-and-drop structural editor supporting Simple Link, Vertical Dropdown, and Mega Menu Grid layout types.
- **Categories Manager:** Nested multi-level categorization editor to structure parent-child pages and link grids.
- **Forms Submissions Reader:** Centralized registry to view, search, inspect, and export lead submissions and job applications.
- **Media Library:** Simple upload manager handling visual assets (images, logos, PDFs) with CDN links.

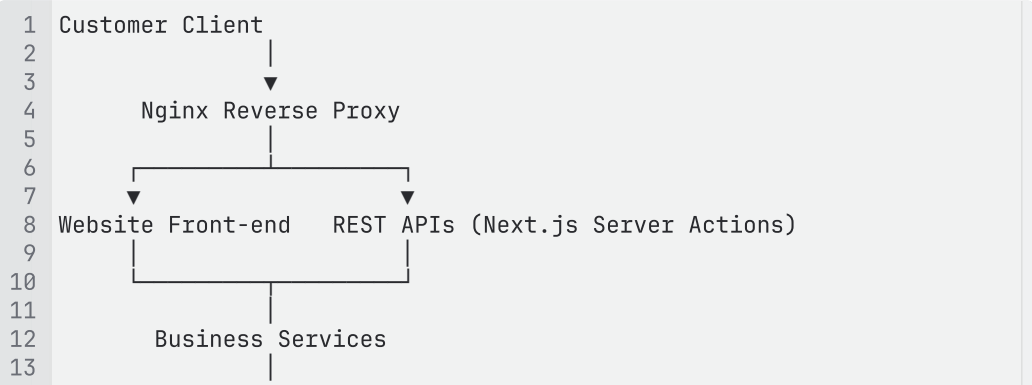
3. Technology Stack

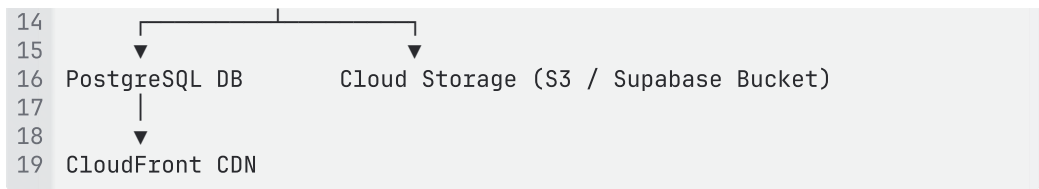
Layer	Technology	Description
Frontend	Next.js 16 (React)	Application framework utilizing App Router & Turbopack compiler.
Backend	Node.js	Runtime server supporting API routing

		and server components.
Language	TypeScript	Strong typing for clean compile safety and validation.
Database	Supabase PostgreSQL	Managed relational database engine.
ORM	Drizzle ORM	High-performance type-safe query builder and database mapper.
Styling	Tailwind CSS & Vanilla CSS	Modern utility-first stylesheet alongside highly-optimized design systems.
Transitions	Framer Motion	Fluid UI micro-animations and slide transitions.
Icons	Lucide React	Rich, high-quality vector icon catalog.
Toast alerts	Sonner	Soft toast notifications for inline user feedback.

4. System Architecture

Below is the conceptual architecture of the ESS India application layout:





5. Delivered Modules

Public Customer Features

Module	Status
Authentication & Sessions	✓ Completed
Dynamic Page Router	✓ Completed
Lead Capture Modals	✓ Completed
Careers & Position Application	✓ Completed
Multi-Behavior Navigation Bar	✓ Completed
Asset Rendering via CDN	✓ Completed
Dynamic SEO Meta injector	✓ Completed

Administrative Features

Module	Status
Admin Dashboard Panel	✓ Completed
Page Wizard Editor	✓ Completed
Navigation & Dropdowns Configurator	✓ Completed
Categories Tree Manager	✓ Completed
Submissions & Careers Inspector	✓ Completed
Media Assets Manager	✓ Completed
Users & Role-Based Access Controls	✓ Completed

6. Third Party Integrations

- **Supabase PostgreSQL:** Managed database and connection pooling layer.
 - **SMTP Integration:** Core outbound mailer server configuration sending email notifications for career application receipts and client lead submissions.
 - **Country Codes List:** Utility database generating country-specific dial code maps dynamically for form entries.
 - **Gemini API KEY:** To extract the text from the JD (/admin/careers)
-

7. Infrastructure

Component	Description
VPS Hosting	Dedicated virtual private server running Linux (Ubuntu 22.04 LTS recommended) to host the application runtime.
Nginx	Serving as the edge reverse proxy handling HTTPS termination, rate limiting, and request forwarding.
Node.js	Production runtime engine hosting the built Next.js server instance.
PM2	Process manager ensuring the Node service automatically restarts on crashes or server reboots.

8. Database Architecture

The system utilizes PostgreSQL, managed and queried via Drizzle ORM. Below is an overview of the key schema tables:

1. users

- Manages administrative personnel and roles.
 - Fields: `id` (UUID, PK), `email` (Varchar, Unique), `role` (Varchar, editor/super_admin/etc.), `fullName` (Varchar), `avatarUrl` (Text), `createdAt`, `updatedAt`.
-

2. pages

- Stores dynamic website pages and configurations.
- Fields: `id` (UUID, PK), `parentId` (UUID), `title` (Varchar), `slug` (Varchar), `fullPath` (Text, Unique), `status` (draft/published), `pageType` (standard/etc.), `categoryId` (UUID), `seoId` (UUID), `createdAt`, `updatedAt`, `publishedAt`.

3. sections

- Contains individual content blocks rendered on dynamic pages.
- Fields: `id` (UUID, PK), `identityHash` (Varchar, Unique), `name` (Varchar), `type` (Varchar), `variant` (Varchar), `contentJson` (JSONB), `styleJson` (JSONB), `usageCount` (Integer), `createdAt`, `updatedAt`.

4. navigation_items

- Configures header menu structures.
- Fields: `id` (UUID, PK), `menuId` (UUID), `parentId` (UUID), `pageId` (UUID), `label` (Varchar), `slug` (Varchar), `url` (Varchar), `megaMenuEnabled` (Boolean), `megaMenuConfig` (JSONB - stores custom dropdown sub-links), `orderIndex` (Integer), `isActive` (Boolean).

5. categories

- Stores hierarchical menu structures and page categorization trees.
- Fields: `id` (UUID, PK), `parentId` (UUID), `pageId` (UUID), `name` (Varchar), `slug` (Varchar), `orderIndex` (Integer), `status` (Varchar), `seoId` (UUID).

6. careers & applications

- Manages job openings and submissions.
- **Careers Fields:** `id` (UUID, PK), `title`, `slug`, `department`, `description`, `type`, `experience`, `location`, `jdUrl`, `budgetRange`, `status`.
- **Applications Fields:** `id` (UUID, PK), `careerId` (UUID, FK), `fullName`, `email`, `phone`, `experience`, `currentCompany`, `noticePeriod`, `resumeUrl`, `coverLetter`, `status`.

7. form_submissions

- Stores lead submissions from customer forms.
 - Fields: `id` (UUID, PK), `name`, `email`, `phone`, `company`, `country`, `message`, `formType` (contact/cta), `pageName`, `pdfUrl`, `status` (new/reviewed), `createdAt`.
-

9. Security Implementation

Authentication & Authorization

- **JWT Access:** Managed API routes authenticated via secure JWT tokens set on HTTP-only cookies.

Input Validation

- **Email Validator:** Strict regular expression checks (`/^[^\s@]+@[^\s@]+\.[^\s@]+$/`) verifying email inputs before submissions are written.
- **Country-Code Mobile Check:** Dynamic, dial-code-aware phone checks mapping Indian mobile numbers (10 digits starting with 6-9), UK numbers, US/Canada numbers, and global boundaries respectively.

Infrastructure Protections

- **PM2 Run isolation:** Running node runtime processes isolated under limited system privilege accounts.
 - **CSRF & CORS Controls:** API endpoints restricted to same-origin requests.
-

10. Deployment Process

Workflow Steps

1. **Push Changes:** Merge code adjustments into the production branch of the GitHub repository.
2. **VPS Access:** SSH into the target production application server:

```
1 | bash
```

```
1 | ssh deploy-user@server-ip-address
```

3. **Pull Updates:** Navigate to the code directory and fetch the latest branch content:

```
1 | bash
```

```
1 | cd /var/www/essindia
2 | git pull origin main
```

4. **Install Dependencies:**

```
1 | bash
```

```
1 | npm install --production
```

5. **Build Application:** Compile and build the optimized production assets:

```
1 | bash
```

```
1 | npm run build
```


6. **Service Restart:** Reload the daemon services via PM2 to apply changes instantly:

```
1 | bash
```

```
1 | pm2 reload essindia
```

11. Operations Guide

Common Administration Commands

- **Check Application Status:**

```
1 | bash
```

```
1 | pm2 status
```

- **View Application Logs (Real-time):**

```
1 | bash
```

```
1 | pm2 logs essindia
```

- **Nginx Reverse Proxy Management:**

```
1 | bash
```

```
1 | sudo systemctl status nginx  
2 | sudo systemctl restart nginx
```

- **Verify Database Connections:** Check PostgreSQL activity via the Supabase dashboard interface.

12. Environment Configuration

The application requires a `.env` file at the root level. Below are the mandatory variables:

```
1 | ini
```

```
1 | # Application Setup  
2 | PORT=5002  
3 | NEXT_PUBLIC_APP_URL=https://essindia.co.in  
4 |  
5 | # Database Config (PostgreSQL / Supabase Connection Pooler)  
6 | DATABASE_URL=postgresql://postgres.xxxx:xxxx@aws-0-xx-  
7 | xxxx.pooler.supabase.com:5432/postgres?sslmode=require  
8 |  
9 | # Authentication Secrets  
10 | NEXTAUTH_SECRET=your-randomly-generated-auth-secret-key-32-chars  
11 |  
12 | # Outbound Mail (SMTP Gateway configuration)  
13 | SMTP_HOST=smtp.mailgun.org  
14 | SMTP_PORT=587  
15 | SMTP_USER=postmaster@essindia.co.in  
16 | SMTP_PASSWORD=your-smtp-credentials-pass
```

```
17 # Storage configuration
18 S3_BUCKET_NAME=essindia-media-bucket
19 AWS_ACCESS_KEY_ID=your-aws-access-key
20 AWS_SECRET_ACCESS_KEY=your-aws-secret-key
```

13. Backup & Recovery

Database Backups

- Managed daily snapshot backups automatically captured in the Supabase management console (retaining up to 7 days of snapshots).
- Manual SQL dumps can be run using the standard tool:

```
1 | bash
```

```
1 | pg_dump -H host -U username -d dbname -f backup.sql
```

Storage Backups

- Set replication rules inside the Supabase storage bucket or AWS S3 destination storage to sync assets across secondary recovery zones.

14. Client Responsibilities

The client is responsible for maintaining ownership, renewals, and payments of the following components to prevent site service interruptions:

1. **GitHub Repository:** Retaining main codebase repositories.
2. **VPS Server Hosting:** Ensuring server resource limits (CPU, memory, storage) meet incoming traffic needs.
3. **Domain Name & DNS Records:** Keeping domain registration active and resolving entries.
4. **Supabase Database Project:** Managed database platform and plans.
5. **SMTP Gateway Provider:** Active subscriptions for dispatching career and submission notifications.

15. Handover Checklist

Handover Item	Status	Comment
Source Code Repository	☐	Access granted to Client GitHub organization.

Database Access	☐	Credentials and ownership transferred.
Production Environment Variables	☐	Handed over via secure channel.
VPS Server Credentials (SSH)	☐	Access credentials created and delivered.
DNS Management (Domain Control)	☐	Confirmed pointing to correct proxy IP.
SMTP Mail Account	☐	Relayed credentials updated.
Technical Documentation	☐	Delivered to the client.
Knowledge Transfer Sessions	☐	Completed with technical team.

16. Support Period

SJ Media Labs provides support covering:

- **Deployment Assistance:** Post-handover server-setup.
 - **Bug Fixes:** Remediations for functional anomalies.
 - **Technical Guidance:** Assistance to internal IT personnel.
 - **Knowledge Transfer:** Explaining layout additions and forms configuration.
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17. Sign Off

By signing below, both parties agree that the project deliverables for **ESS India Portal** have been successfully completed, tested, and handed over.

Client Representative

Name: _____

Signature: _____

Date: _____

SJ Media Labs Representative

Name: _____

Signature: _____

Date: _____

Thank You

ESS India Project Handover

Prepared by SJ Media Labs

"Building scalable digital products with enterprise-grade engineering."